27 (a) Write the number 0.000 075 47 in standard form.

Move the decimal 5 places right.

7.547×10^{-5}

(1)

1 mark

Compensate with a power of −5

(b) Write 3.42×10^4 as an ordinary number.

$3.42 \times 10\,000 = 34\,200$

34200

(1)

1 mark

Move the decimal 4 places right

(c) Work out $\dfrac{2.3 \times 10^4 \times 6.7 \times 10^3}{5 \times 10^{-8}}$

Numerator: $23\,000 \times 6700 = 154\,100\,000$

1 mark

Carry out a correct intermediate calculation

$$\frac{154\,100\,000}{5 \times 10^{-8}} = 3.082 \times 10^{15}$$

1 mark

Correct final value

3.082×10^{15}

(2)

(Total for Question 27 is 4 marks)